

KETAN KATORE

SENIOR SOFTWARE ENGINEER | FULL-STACK | BACKEND | REAL-TIME SYSTEMS

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PROFESSIONAL SUMMARY

Senior Software Engineer with 6+ years of experience designing, developing, and scaling full-stack, backend, and real-time applications. Strong across TypeScript, JavaScript, Node.js, Angular, React, Next.js, REST APIs, MongoDB, MySQL, Redis, and PostgreSQL, with experience owning features from technical design through production release and support. Deep expertise in WebRTC and real-time media, alongside microservices, distributed systems, performance optimization, production debugging, automated testing, and enterprise application development.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), HTML5, CSS3, SASS/LESS

Frontend: Angular, React, Next.js, RxJS, UI Components, State Management & Synchronization

Backend: Node.js, Express.js, Sails.js, REST APIs, Microservices, JWT, OAuth2

Real-Time & Communications: WebRTC, Janus, SFU/MCU, Socket.io, SIP, RTP/RTCP, SDP, ICE, STUN, TURN, TCP/UDP, Real-Time Signaling

Media & Video: FreeSWITCH, Citrix Workspace SDK, Banuba SDK, Google MediaPipe, MediaStream & Device Management

Databases & Data: MongoDB, MySQL, PostgreSQL, Redis, Aggregation, Sharding, Schema Design, Query Optimization

Performance & Reliability: Network Resilience, ICE Restart, TURN Fallback, Jitter/Packet-Loss Analysis, Bandwidth Adaptation, getStats(), Production Debugging

Engineering & Delivery: System Design & Architecture, Unit Testing, Integration/E2E Testing, Code Reviews, Technical Mentoring, CI/CD, Agile/Scrum, Technical Documentation, Git, GitHub, JIRA

PROFESSIONAL EXPERIENCE

Spring Computing Technologies Pvt. Ltd. | Senior Software Engineer / Senior WebRTC Engineer

Jan 2023 – Present

- Contribute to a global communication platform serving approximately 20,000 users per day as part of a 4–5 engineer team, owning key real-time media and reliability components.
- Own end-to-end delivery of enterprise media features, taking responsibility from technical investigation and implementation through testing, production release, and support.
- Own client-side media strategy across web and VDI/Citrix environments, designing solutions for reliable performance, device recovery, and media processing.
- Improve WebRTC resilience through ICE Restart, STUN/TURN, relay fallback, adaptive bitrate/framerate, substream selection, CPU-aware controls, and real-time telemetry.
- Integrate Citrix Workspace SDK for VDI media offloading and Banuba and Google MediaPipe for client-side virtual backgrounds and real-time video effects.
- Develop media and resource controls that dynamically adjust video resolution and stream configuration based on real-time client/server conditions.
- Investigate and resolve complex production issues involving audio/video synchronization, network jitter, packet loss, device handling, media negotiation, and FreeSWITCH recording layouts.
- Participate in code reviews, technical mentoring, documentation, and release cycles from pre-alpha through General Availability while independently owning assigned technical areas.

Spring Computing Technologies Pvt. Ltd. | Software Engineer

Feb 2021 – Dec 2022

- Engineered Angular frontend and Node.js backend capabilities for an enterprise Unified Communications and business telephony platform.
- Designed and implemented calling, messaging, and collaboration workflows including call transfer, call parking, ad-hoc conferencing, device binding, presence tracking, real-time chat, SMS, and file sharing.
- Developed backend services and real-time signaling integrations while synchronizing browser application state with the communication layer.
- Worked with scalable service and data architectures, including Redis-backed real-time communication patterns and database-driven application workflows.
- Owned high-priority production fixes, code reviews, and automated testing practices to improve stability across core messaging and calling modules.

Humation Limited | Associate Software Engineer

Jul 2020 – Jan 2021

- Led the overhaul of a legacy Node.js backend into a modular architecture using MongoDB, improving maintainability and release speed.
- Implemented secure authentication and API functionality and optimized response times for low-latency financial-grade applications.

OneGreenDiary Software Pvt. Ltd. | Associate Software Engineer

Jan 2020 – Jun 2020

- Delivered end-to-end features using Angular, Node.js, Sails.js, and MySQL across frontend, backend, database, and integration layers.
- Engineered automated data-processing and reporting modules to streamline business analytics and reporting for enterprise clients.

SELECTED PROJECTS

Intermedia AnyMeeting — High-Scale WebRTC Video Conferencing | Senior Software Engineer / WebRTC

- Engineered ICE Restart and TURN/Relay fallback logic to improve call resilience and recovery across restrictive network conditions.
- Overhauled hardware and media-device handling for fast device recovery and graceful peripheral hot-swapping with real-time UI synchronization.
- Integrated Citrix Workspace SDK to offload WebRTC media processing to local endpoints and reduce server-side processing overhead.
- Led integration of Banuba and Google MediaPipe for client-side virtual backgrounds and real-time video effects.
- Implemented adaptive bitrate/framerate, substream handling, and CPU-aware media controls to maintain video quality and stability under changing network and resource conditions.
- Resolved complex production issues involving audio/video synchronization, network jitter, packet loss, device behavior, and FreeSWITCH recording layouts.

Zultys ZAC — Unified Communications & Business Telephony | WebRTC & Full-Stack Developer

- Developed Node.js middleware for Janus Media Server, managing room lifecycles and ad-hoc conferencing for thousands of users.
- Implemented server-side load distribution to enable media infrastructure to scale horizontally across multiple instances.
- Integrated enterprise PBX capabilities including Call Park, Transfer, Device Binding, and conferencing into the web client.
- Built Angular UI for a high-performance Video Preview Widget with hardware and media-stream validation before session entry.

NFT Marketplace & Artist Appreciation Platform | Full-Stack Developer

- Developed Angular, Node.js, MongoDB, and Socket.io functionality for real-time bidding and high-concurrency auction synchronization.
- Built secure multi-currency transaction, wallet, and withdrawal workflows supporting Bitcoin, Ethereum, and Litecoin through third-party APIs.
- Led backend re-engineering into a modular architecture to improve maintainability and scalability.

Cloud-Based Point of Sale Ecosystem | Full-Stack Developer

- Architected an automated reporting server generating daily, weekly, and monthly sales analytics in Excel and PDF formats.
- Built licensing, feature-toggle, client onboarding, FAQ, content-management, and REST API functionality using Angular, Node.js, Sails.js, and MySQL.
- Delivered checkout-system integration with low-latency synchronization between client and application-server layers.

Farmer-to-Consumer Direct Marketplace | Backend Developer

- Designed and deployed Node.js/Express/MySQL backend infrastructure for a mobile-first group-booking platform connecting farmers directly to consumers.
- Built dynamic combo management, real-time demand generation, area-based outreach, and high-volume transaction/inventory data models.
- Engineered WhatsApp deep-linking and sharing functionality to support community-driven group bookings.

EDUCATION

M.Sc. Scientific Computing | Savitribai Phule Pune University | 2020